Proposal title: California Wildfire Risk and Impact Analysis
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1 INTRODUCTION

In recent years California has faced increasing wildfire frequency and intensity due to climate change, drought, and expanding human development across the state. These fires threaten lives, property, ecosystems, and at-risk populations due to reduced air quality. Using machine learning models for wildfire prediction and visualization is critical for safeguarding communities, optimizing emergency response, and informing land-use and insurance policies. These data-driven, real-time analytical models assist in creating better wildfire preparedness and mitigation strategies.

2 PROBLEM DEFINITION

Current wildfire risk maps are largely static. These models are usually based on historical fire, vegetation, and climate data—making them poorly suited for real-time decision-making or dynamic risk assessment. This limits the ability of emergency agencies and residents to anticipate and respond effectively.

3 LITERATURE SURVEY

Relevant fire data is often collected through remote sensing (Smith et al, 2019); they emphasize the need for utilizing a mix of data sources to overcome the limitations of each sensor. The limitation of this study is in its focus on dryland areas, whereas we hope to focus on all of California. Remote sensing has been combined with deep learning to assess wildfire risk (Xu et al, 2025). Xu's work demonstrates the usefulness of variable selection and ConvLSTM on wildfire data with multicollinearity, however their work focuses on the risk and starts of fires, not how they spread. The same author worked on an overview of wildfire prediction methods (Xu et al, 2024). Xu emphasizes elements of effective wildfire prediction such as multi-modal inputs and calibration techniques, providing ideas for a better model approach. Fathima et al. (2021) provides game theoretic approaches utilizing Shapley values derived from linear regression to identify relevant features such as wind speed and ignition probability. This paper uses Adversarially Learned Anomaly Detection, a GAN that creates high accuracy of fire spread prediction, but is limited as it does not consider continuous spread. We will expand the study by incorporating temporal evolving wind and fuel data. Studying continuous fire spread, Bhowmik et al. (2023) focused on meteorological data (atop environmental and vegetative data). They provide an U-Convolutional LTSM neural network that works with spatiotemporal data, such as continuous fire spread. We will improve the study's limited variables using satellite data that gives daily fire spread. Huot et al. (2022) focuses on ConvRNNs and ConvLSTM with autoregression to predict fire spread. Their modeling sequence method sequences fire spread over short periods of time to predict fire spread, but they focus on planet wide google earth data; our focus is on California, and our smaller scale could result in a stronger model.

The current standard approach to applying AI in wildfire modeling relies on three main factors: fuel availability (fFA), fuel flammability (fFF), and human suppression (fHS). Datasets used within these groups may vary, they consistently serve as the foundation linking environmental and human drivers to fire impact through machine learning models. In Li et al. (2024), a hybrid machine learning framework that incorporates an attention mechanism, similar to those used in LLMs, captures correlations between fuel flammability and fire scale. This significantly improved predictive performance compared to existing models (Li et al., 2024). With these approaches, studies like Reid et al. (2016) demonstrate the need for accurate and timely fire spread prediction by linking smoke exposure to respiratory morbidity for California residents, but do not offer a solution.

Malik et al. (2021), uses random forests to predict wildfire ignition; the authors used a labeling method of positive and negative targets that may assist our model variable labeling. This study's use of random forest is unideal given the model's inherent lack of interpretability, which opposes our project's main goal of interpretability. Parvar et al. (2024), focuses on fire ignition risks instead of spread or impact. Topography, human factors, climate, and vegetation, with additional sub-criteria, were used to determine high-fire-risk areas in Golestan Forest. This was paired with

GIS-based Forest Fire Risk (FFR) maps to integrate complex layers of information into an accessible visualization, allowing easy interpretation.

Particulate matter is a common way to measure wildfire spread. Aguilera et al. (2022), measures particles <2.5 micrometers in diameter (PM2.5) throughout California, demonstrating the ability to create statewide exposure data with multiple variables (wind, elevation, land type), and shows a link between fire spread and air quality, but is limited due to outdated data. Finite particular matter has been found to affect different populations with varying levels of severity (Chunga Pizarro et al., 2024). This study used time series methods to estimate when PM2.5 concentration became unsafe for farmworkers in danger of wildfires, similar to methods we plan on using to track fire change over time. However, the study is solely based on data gathered after smoke is detected through sensors, without anticipating live fire spread updates. Other wildfire air quality studies focus on particulate matter concentration including emissions, meteorology, and dust emissions (Carreras-Sospedra et al., 2024). The study was limited from August to October 2020, reducing generalizability, but incorporates layers, similar to our GIS integration goals.

Shadrin et al. (2024) predicts the spread of wildfire using a neural network factoring in a time range of 1 to 5 days as the forecast horizon. The study considers the importance of feature selection from climate and environmental data. We can improve this by obtaining more detailed weather forecast data. Hu et al. (2024) uses deep neural networks along with long term memory networks to predict wildfire likelihood, cause, and size. While the methodology utilizes a multi-faceted prediction approach, the data only goes to 2018 and doesn't integrate parameters such as climate data. Similarly, Castrejon et al. (2023) uses a Multi-Layer perceptron model trained with NASA remote sensing data. Similar to our visualization goals, this study created 3D visualizations from 2D satellite images. Where Castrejon's data imbalance between fires and non-fires is a limitation, Langford et al. (2018) benefits from imbalanced datasets. Langford uses deep neural nets to map wildfires in Alaska, which provides model insights but lacks transferability to California. Pham et al. (2020) compares various bayesian classification algorithms to tree based methods, and deals with visualizing various layers incorporated into fire prediction. However, the study's scale was smaller than California (Pu Mat National Park).

4 PROPOSED METHOD

Our goal is to represent wildfire risk and impact areas across California through machine learning classification models and an interactive dashboard, in addition to GIS data mapping. This will be done through analysis of CAL FIRE, climate, air quality, and demographics data. The outcome for this project will be a spatial-temporal model that allows for real-time wildfire risk assessment through quarterly measurements of data across California.

4.1 GIS Data Acquisition and Data Preparation

The GIS component of this project focused on constructing a comprehensive, spatially standardized geodatabase of wildfire-related datasets for the state of California. Data were sourced from CAL FIRE, state boundary layers, historic fire perimeters, topography, vegetation, demographics, and air-quality repositories. Each dataset was validated for spatial accuracy, projected to the NAD 1983 California (Teale) Albers coordinate system (EPSG 3310), and aligned within a consistent 1-mile statewide grid. This grid serves as the spatial index for all subsequent analyses, ensuring consistent geographic reference across all data layers (Chuvieco et al., 2014; Parvar et al., 2024).

After acquisition, all datasets were compiled into a unified ArcGIS Pro workspace. Feature classes were re-projected, standardized, and assigned GlobalIDs to maintain stable unique identifiers across transformations and joins. These GlobalIDs also enable bidirectional data linkage—allowing results generated from machine learning models to be accurately reprojected and reintegrated into the GIS environment for spatial analysis and visualization. Using ArcPy, spatial joins and intersections were performed to merge multiple thematic layers, capturing spatial relationships among climate, environmental, and human variables influencing wildfire occurrence and spread (Li et al., 2024; Huot et al., 2022).

The finalized CalFire_Master_DataSet_C represents the complete geospatial deliverable provided to the Data Acquisition and Preparation Group. It includes all core spatial layers, attribute documentation, and schema optimizations necessary for downstream processes. This master dataset will serve as the foundation for advanced data cleaning, data extraction, feature engineering, and ultimately for machine learning modeling, with seamless reintegration into GIS for spatial-temporal prediction, mapping, and decision-support visualization (Bhowmik et al., 2023; Xu et al., 2025).

This finalized geospatial dataset can then be used in addition to temporal datasets including average pm2.5 data from EPA records and air speed, humidity, soil temperature, dewpoint, air temp, precipitation, and vapor pressure records from California Irrigation Management Information System (CIMIS) stations around the state to create a spacial-temporal dataset. The original geospatial dataset has 368 columns with layers including demographics, roadway, vegetation, environmental hazard, and fire perimeter/risk data. To implement the finalized dataset's temporal aspect, we decided to use quarterly averages of sensor reports to reduce computational load training the model. There are over 165,000 rows in the geospatial dataset, and the model will be trained on data from 2000 to 2025, leading to over 16,500,000 data points using quarterly averages.

After converting the temporal sensor recordings for both EPA and CIMIS data to quarterly reports for each station, the spatial data must be connected to the sensors; this is done by stacking the spatial data for each quarterly date, then calculating the euclidean distance of the specific grid section's (row's) longitude and latitude to all existing sensors on that date. For the closest station by longitude/latitude, that station's quarterly report is attached to the section. From this, we take the original dataset's fire alarm date to create a binary No Fire (0)/Fire (1) attribute if the alarm date was recorded in the respective quarter, and attach the fire acres value to each respective positive fire binary.

We will be using a random forest classifier as well as logistic regression to establish a baseline prediction level with traditional machine learning methods as a reference. For these models, the derived binary fire variable, created in the data preparation phase, is used as the target. The latitude and longitude along with relative percentage metrics are converted to a min and max scale from zero to one while the other metrics are converted to a z-score scale due to the appearance of outliers skewing the data. The remaining categorical variables are prepared for modeling using one-hot encoding.

4.2 Model Evaluation

To help evaluate these models, stratified k-fold validation with five folds will be used to ensure even distribution of the binary target variable. This is due to having an imbalanced dataset since less than a third of a percent contain a fire. Both of these models will have accuracy, precision, recall, F1 and AUC computed for each fold. These will then be averaged to determine the overall metrics in how effective the models are. In addition, for each of the folds the models will be trained and then saved to a pickle file. The predictions will also be tied to the test split for the corresponding fold. This allows for greater flexibility for data visualization.

4.3 Visualization

To display our model, we initially planned to utilize the Tabpy library in Python to view predictions in Tableau. However, Tabpy does not work with the student version of Tableau that we have access to with our student accounts, Tabpy works with Tableau Desktop, which we only had a two week trial access to use. Thus, we pivoted to using Javascript and the D3 library so we could utilize bespoke data visualization. Currently we have a map of California and lines according to zip code, the graph itself is draggable and zoomable, and the fires are plotted based on their latitude and longitude and are using a small sample of our dataset that is being used for the machine learning process (the visualization dataset has about ten thousand rows instead of the full dataset which is over six gigabytes

in size). We have also explored various designs for the overall visualization dashboard. The final design can be found in *Appendix 7.1: Visualization dashboard*.

We have chosen to have the fire bubble size and color be based on Fire Predicted Probability, fires that produced lower FirePredicted_Prob values are smaller and white while fires that produced higher FirePredicted_Prob values are larger and red. The visualization operates by loading in a csv file with appropriate column names (e.g.Census2020_INTPTLAT20, Census2020_INTPTLON20, and FirePredicted_Prob) and then click a “load” button, there is also an option to load in and display zip code lines with a json file of appropriate zip coordinates. We have implemented a few sliders to help clarify what kind of fires we are looking for and how to better see them, such as a slider for fire probability that acts as a filter (a higher fire probability value on the slider filters out fires with lower values), a slider for fire bubble opacity, and a slider for zip code line opacity. While testing the visualization with thousands of rows we noticed that the fire bubbles can overlap and make the individual fires difficult to see. To fix this we implemented a variable bubble size controlled by adaptive sampling based on zoom level on both the server and client end. The fire bubbles are still relatively sized to one another (smaller fire probability values have smaller bubbles), but visualizing fires with higher fire probability values quickly becomes messy and greatly benefits from the size slider. This logic will also be useful in the future when the machine learning process is completed.

5 Evaluation

5.1 Predictive Modeling

After testing the baseline predictive models, support vector machines were not as viable as expected; this was due to unexpected length of training time for the model, taking over six days to perform the first stratified fold of training. With this, the results of the model would not be finished by the due date of the project; to counteract this, logistic regression was chosen instead of SVM to get baseline predictions for the dataset. To ensure quicker training and best results of baseline models, three datasets were trained: annual, biannual, and quarterly datasets were created.

Our predictive results from logistic regression and random forest both show success with predicting wildfires across the state. With the annual dataset, logistic regression had an accuracy of 0.9412 and an AUC score of .92; this model struggled primarily with precision (0.1287), but additionally had issues with recall (0.7620), and f1 scores (0.2202). In comparison to the logistic regression model, the yearly random forest model had an accuracy of 0.9998 alongside AUC, precision, recall, and f1 scores of .9999, .9999, .9906, and .9953 respectively. Because of such a large discrepancy, random forest was the only model that continued to be tested for the biannual and quarterly datasets. Decreasing time between temporal measurements slightly changes model performance; both the quarterly and biannual models retain accuracy scores of .9999, AUC scores of .9999, and precision scores of 1.0; recall and f1 scores are the only major difference between the three different models, with the biannual model having scores of .9930 and .9965 and the quarterly model having scores of .9849 and .9924 respectively.

We evaluated a neural network as a potential model for quarterly fire prediction. Initial model performance reflected the issue of extreme class imbalance in the dataset. Although accuracy reached 0.991, both precision and recall were effectively zero due to over 99% of grid cells not containing a fire. The model learned that always predicting “no fire” minimized the loss, which completely defeated the model’s purpose. To address this, we applied BCEWithLogitsLoss with a weighted positive class so the model would penalize false negatives more heavily. However, an excessively large weight caused the model to swing in the opposite direction and predict all positives. We therefore tuned the pos_weight parameter carefully and oversampled fire events in the training set to stabilize learning. These adjustments produced meaningful probability outputs, though they clustered tightly within a narrow range, requiring careful threshold tuning to balance accuracy, precision, and recall. The best configuration achieved an accuracy of 0.723, recall of 0.342, and precision of 0.241. Increasing the model complexity or training duration did

not improve results—the neural network simply lacked enough informative signal to distinguish fire from no-fire conditions effectively, even though the Random Forest model performed substantially better on this task.

In comparison to previous studies, these results show that random forest had better prediction than the neural network. A random forest is a group of decision trees that were trained on a dataset; after many epochs the trees converge to classify these data points to fire or not fire. This process creates a model that is dependent on a dataset, potentially overfitting to the data available. In this project's scope, this limitation is fine because the model performed much better than the alternative neural networks. In comparison to random forests, neural networks try to find the weights that show the pattern between the training data and target. These models learn the general patterns, showing what data would lead to a positive fire binary. This is much less prone to overfitting, but there are too many other variables that cannot be captured within the data in real life, leading to a worse model. The patterns within the data's predictor variables are not as apparent when predicting the fire binary, leading to random forest offering higher predictive power for the given data, as it does not have many abstractable patterns.

5.2 GIS Analysis and Predictive Mapping

The GIS component of the California Wild Fire Risks and Impacts Analysis project built a consistent, statewide system for turning raw spatial data into monthly wildfire-risk predictors. All geoprocessing was carried out in ArcGIS Pro 3.5 with ArcPy used to automate dataset preparation, cleaning, and assembly. File Geodatabases stored intermediate layers and standardized datasets, supporting reliable spatial joins and other operations. Point and polygon layers were projected into a common coordinate system, checked for geometry problems, and aligned by month so each ignition record matched the correct spatial unit. This produced a single, coherent framework where vegetation, drought conditions, terrain characteristics, weather summaries, and past fire activity were expressed uniformly across California, allowing both the neural-network model and the Random Forest model to learn from patterns tied to real landscapes.

Environmental and administrative boundary layers were processed through attribute checks, spatial joins, and polygon-level summaries to create a shared set of predictors for each analysis unit. These clean, standardized predictors were merged with ignition-based labels and organized into feature matrices for model training. After predictions were generated statewide, the outputs from the neural-network and Random Forest models were written back to their corresponding polygons and stored in a visualization geodatabase. This allowed the results to be examined directly on the landscape, making it possible to compare how the two modeling approaches handled regional patterns, understand where they agreed or differed, and relate predicted ignition risk to established fire-history trends.

The combined prediction layers displayed clear spatial structure, highlighting areas with a long history of higher ignition likelihood and revealing how risk tends to cluster across the state (Appendix 2). The visualization geodatabase supported the production of continuous probability surfaces and threshold-based high-risk maps, providing practical tools for evaluating environmental drivers and understanding regional tendencies. The workflow processed roughly 165,000 polygon-month records and is capable of scaling as more detailed datasets are added. Overall, the GIS system delivered a stable and flexible spatial pipeline for wildfire-risk modeling using both neural-network and Random Forest methods, while leaving room for future enhancements such as finer-resolution environmental inputs, improved temporal alignment, and more detailed spatiotemporal analysis.

5.3 Visualization

Our visualization was the subject of multiple optimization attempts. Initially we used D3's json function to load a url of zip codes, and our second iteration made use of a locally loaded json file to display the zip code lines. However the loading was discovered not to be the issue, each zip code line is a separate path in SVG. When the visualization is

zoomed in or out, every polygon of every element of the json file is redone because D3 is applying a transform scaling, which regardless of loading method results in visual lag. We attempted various methods to work around this but we were not able to implement them by our deadline. The first attempt was to only display zip codes at a sufficient zoom level but this only moved the efficiency bottleneck to another zoom level. We also attempted to hide zip code lines while zooming in and then redisplay them but this created a jarring pop-in effect and a flickering of lines. Our final visualization, firebug, needs to a local server to be run, utilizing docker to create a local server, utilizing Typescript, React, Python, Flask, and Uvicorn; this model no longer shows zips, reducing lag while showing major features within the state.

Utilizing javascript ensures that the visualization is lightweight, it requires an internet connection to load the local docker server. Once loaded, all interaction and rendering are done on the client's computer, which makes our project easily accessible across browsers. The model allows for swapping between historical fires with the ability to filter by attribute, along with predicted fire-prone areas with thresholding. Additionally, a temporal bar graph is shown above the map, giving fire-count by month, visualizing fire severity for each month's data.

6 Discussion and Conclusion

The goal of this project was to utilize machine learning to improve upon past wildfire prediction models of California. This was done by applying a spatial-temporal grid across the state, using a GIS-based spatial dataset with temporal sensor data, allowing for future prediction based on future sensor data. To predict the fire binary variable, random forest, logistic regression and a neural network were used. Based on previous research, it was expected that the neural network would give the highest accuracy, but our models differed, with the random forest model showing the highest rate of success. This is most probably due to a combination of overfitting of the random forest in addition to the neural network not extracting observable patterns because of variables of the environment not captured within the data. Random forest was chosen as the model to be used for visualization even though it lacks interpretability due to its overwhelming success in accuracy, precision, and recall.

These predictions were applied into both GIS predictive mapping high-risk areas and interactive visualizations using a combination of Typescript, React, Python, Flask, and Uvicorn. Within GIS, the prediction mapping created visualizations of areas of high ignition rates. The interactive visualization of California maps historical and predicted fire zones across the state, showing monthly trend data with a temporal bar graph above the map. Wildfires can be filtered by year, cause, and agency responsible; thresholding can be applied to predicted fire areas to see only higher-risk areas.

All team members have contributed a similar amount of effort.

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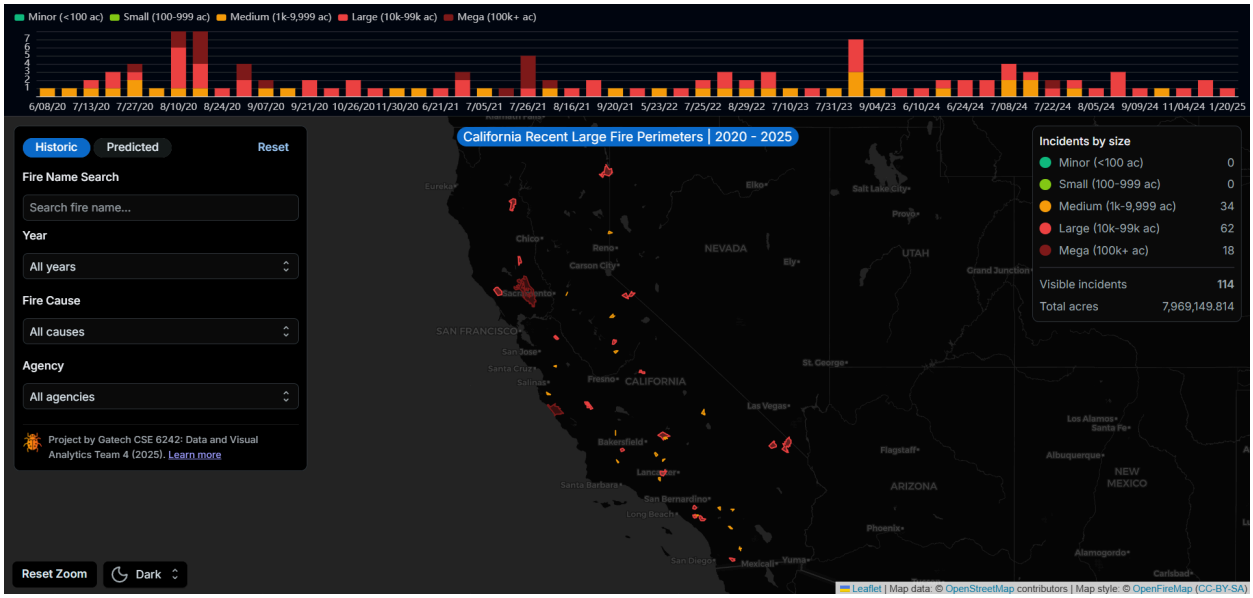
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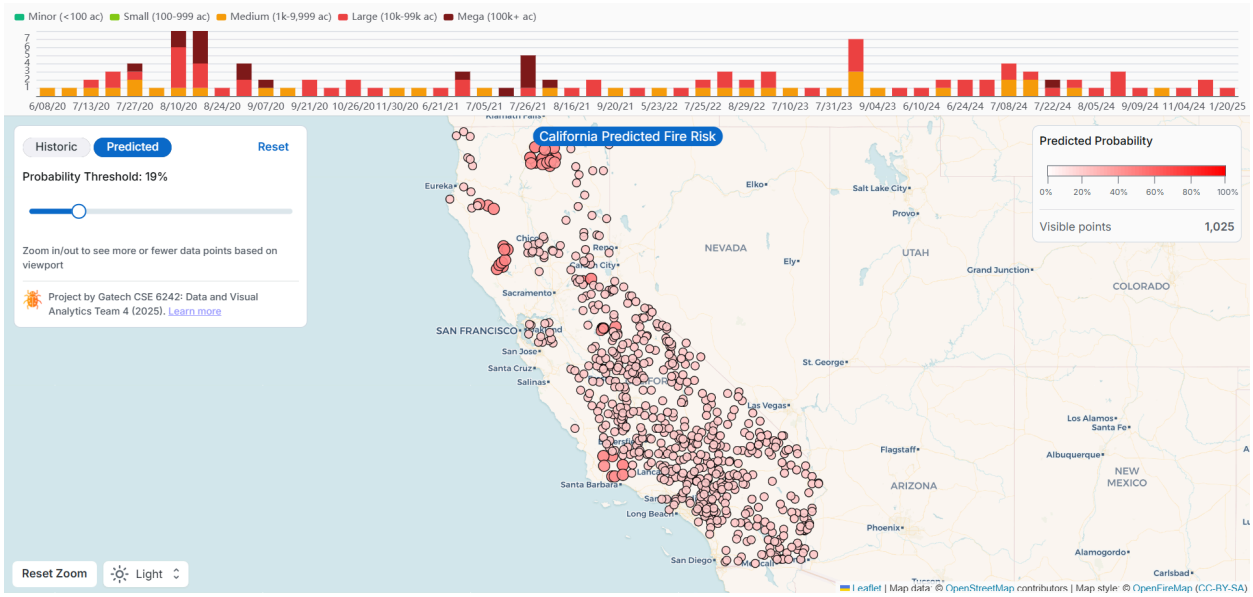
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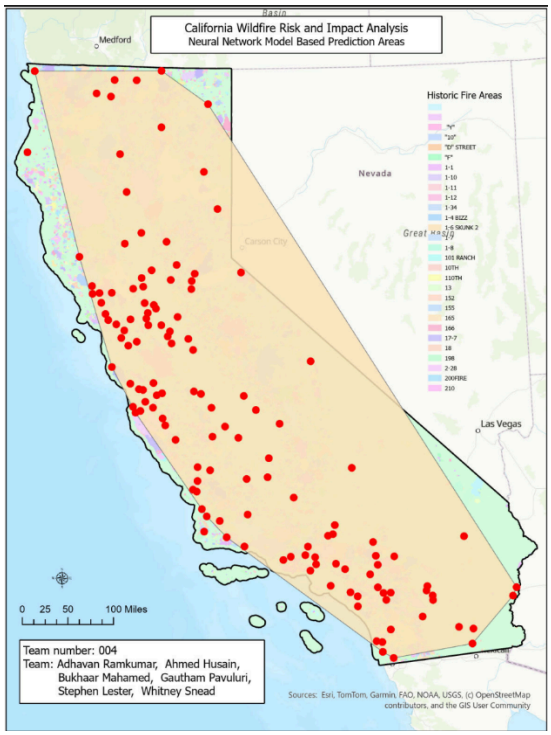
7 APPENDICES

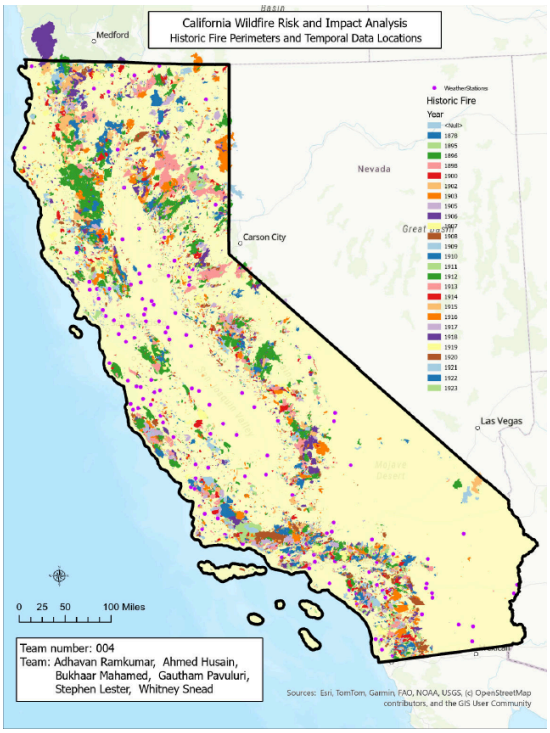
7.1 Visualization dashboard

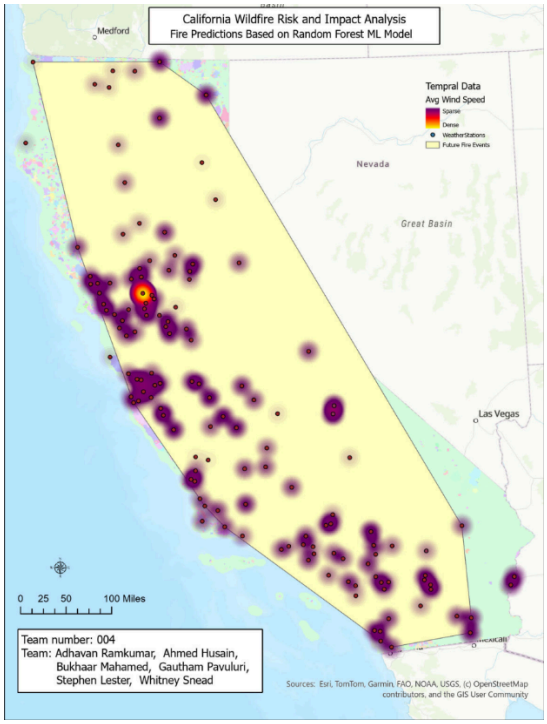


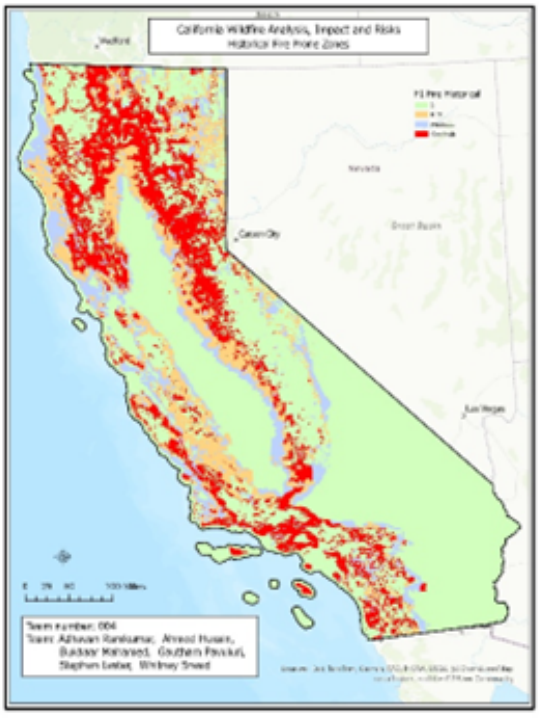


7.2 GIS Prediction Outputs



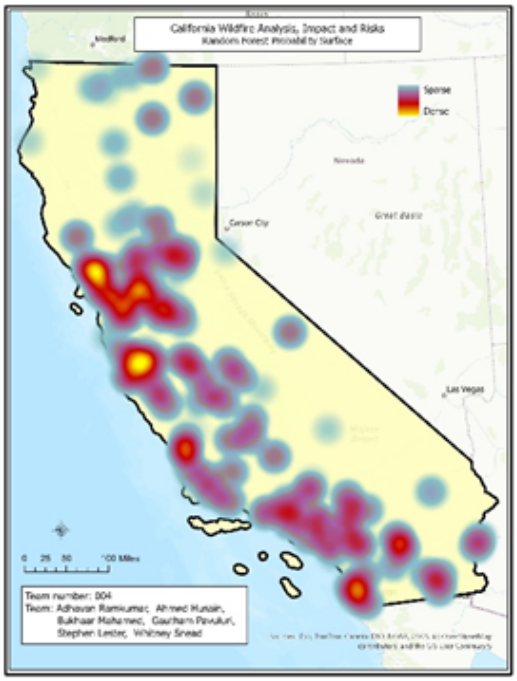


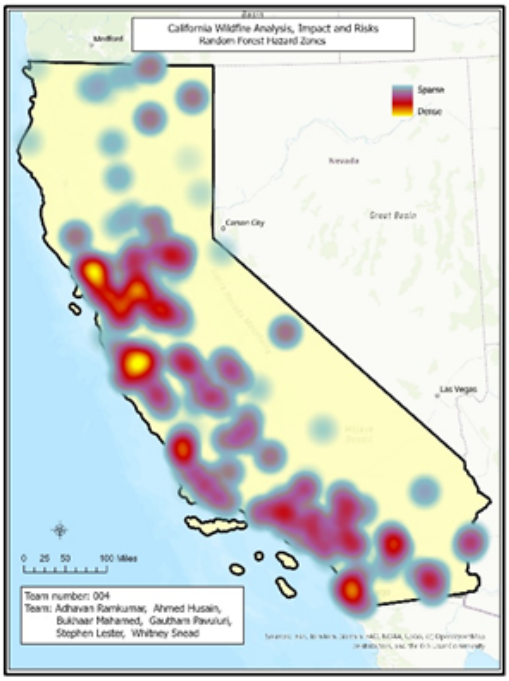


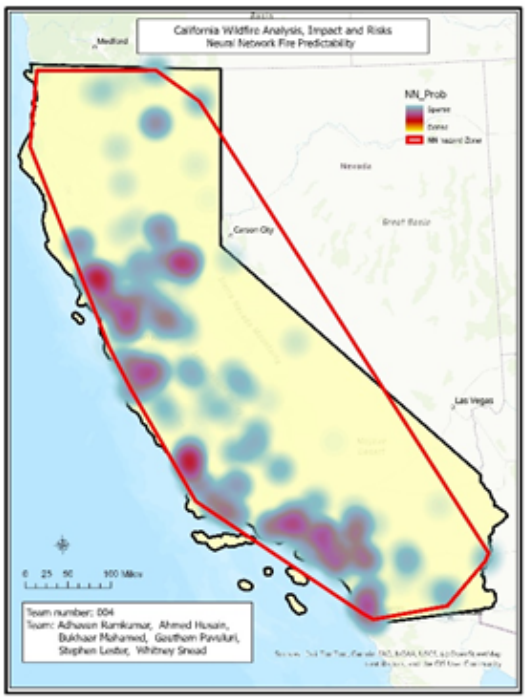


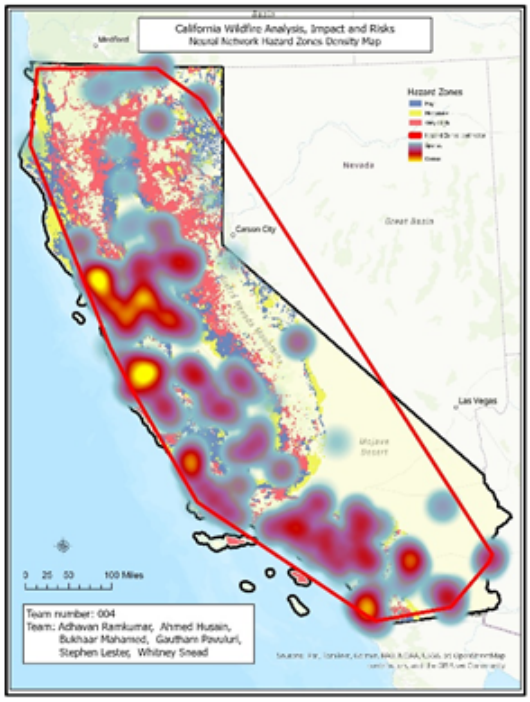

California Wildfire Analysis, Impact and Risks
Historical Weather Stations and Temporal Data Collection Points
Team number: 004
Team: Adhavan Ramkumar, Ahmed Husain, Bukhaar Mahamed, Gautham Pavuluri, Stephen Lester, Whitney Snead


California Wildfire Analysis, Impact and Risks
ML Training Points
Team number: 004
Team: Adhavan Ramkumar, Ahmed Husain, Bukhaar Mahamed, Gautham Pavuluri, Stephen Lester, Whitney Snead


California Wildfire Analysis, Impact and Risks
Random Forest Probability Surface
Sparse
Dense
Team number: 004
Team: Adhavan Ramkumar, Ahmed Husain, Bukhaar Mahamed, Gautham Pavuluri, Stephen Lester, Whitney Snead


California Wildfire Analysis, Impact and Risks
Random Forest Hazard Zones
Sparse
Dense
Team number: 004
Team: Adhavan Ramkumar, Ahmed Husain, Bukhaar Mahamed, Gautham Pavuluri, Stephen Lester, Whitney Snead


California Wildfire Analysis, Impact and Risks
Neural Network Fire Predictability
NN_Prob
Medford
Nevada
Carson City
Great Basin
Las Vegas
Mojave Desert
0 25 50 100 Miles
Team number: 004
Team: Adhavan Ramkumar, Ahmed Husain, Bukhaar Mahamed, Gautham Pavuluri, Stephen Lester, Whitney Snead


California Wildfire Analysis, Impact and Risks
Neural Network Hazard Zones Density Map
Hazard Zones
Medford
Nevada
Carson City
Great Basin
Las Vegas
Mojave Desert
0 25 50 100 Miles
Team number: 004
Team: Adhavan Ramkumar, Ahmed Husain, Bukhaar Mahamed, Gautham Pavuluri, Stephen Lester, Whitney Snead


California Wildfire Analysis, Impact and Risks
NN vs RD Prediction Comparison of Hazard Areas
Medford
Nevada
Great Basin
Las Vegas
0 25 50 100 Miles
Team number: 004
Team: Adhavan Ramkumar, Ahmed Husain, Bukhaar Mahamed, Gautham Pavuluri, Stephen Lester, Whitney Snead


California Wildfire Analysis, Impact and Risks
NN Error Analysis
Medford
Nevada
Carson City
Great Basin
Las Vegas
Team number: 004
Team: Adhavan Ramkumar, Ahmed Husain, Bukhaar Mahamed, Gautham Pavuluri, Stephen Lester, Whitney Snead